

EMBEDDED SYSTEMS

In Automobiles



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In today's world, most electronic devices are based on embedded systems. From home appliances like microwaves and washing machines to entertainment and security systems, embedded systems have found a way into every field.

What an embedded system is

An embedded system is a computer system with a dedicated function within a large mechanical or electrical system, often with real-time computing constraints. Key characteristics of an embedded system are:

1. Sophisticated functionality
2. Real-time operation
3. Low manufacturing cost
4. Low power

Mechanical systems in automobiles have largely been replaced by electronic systems. Today, the automobile industry is making great use of embedded systems. Ranging from wiper controls to complex anti-lock brake controls and air bags, embedded systems have gained overall control of automobiles.

Automobiles that are built around micro-controllers, digital signal processors or both processors are commonly called electronic control units. Many luxury vehicles have come up with a large number of embedded controllers. The first embedded system based

automobile, Volkswagen, came in 1968.

Some current trends of embedded systems in automobiles include air bags, event data recorders, anti-lock brake systems, cruise control, rain-sensing wipers, emission control, traction control, automatic parking, in-vehicle entertainment, back-up collision sensors, navigation systems and tire-pressure monitors.

Airbags

The airbag system is an important safety device that provides extra protection against head-on crash for the passengers, by giving a soft surface to land on. This system works on commands from the airbag control unit, which has a micro-controller. The controller gets power from the battery.

If the collision sensor detects an accident, a signal is sent to the airbag control unit and it is processed by the airbag control unit to determine severity of the impact. If airbag deployment is necessary, the airbag control unit sends a signal to initiate airbag inflators. Inflators are activated through an igniter, causing a chemical reaction that emits nitrogen gas, resulting in the deployment of the airbag cushion.

An occupant detection system is used to determine if a person is seated in the passenger seat and if he or she is of adequate size to be protected in the event of deployment of the passenger seat airbag. It measures the weight of the passenger to determine if the corresponding airbag should deploy.

In 2012-13, a new type of occupant detection system called electrostatic capacitance sensor was implemented. This system does not use weight to determine whether to turn the occupant detection system on or off.

Electrostatic capacitance represents a material's capability of storing an electrical charge. When someone is seated or when

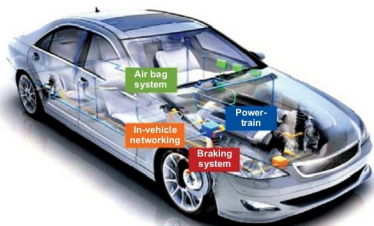


Fig. 1: An automobile with embedded systems